Numerical semigroup tree: type-representation

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Numerical semigroups

A *numerical semigroup* H is a subset of $\mathbb{N}$ closed to addition, $0 \in H$ and $\mathbb{N} \setminus H$ is finite.

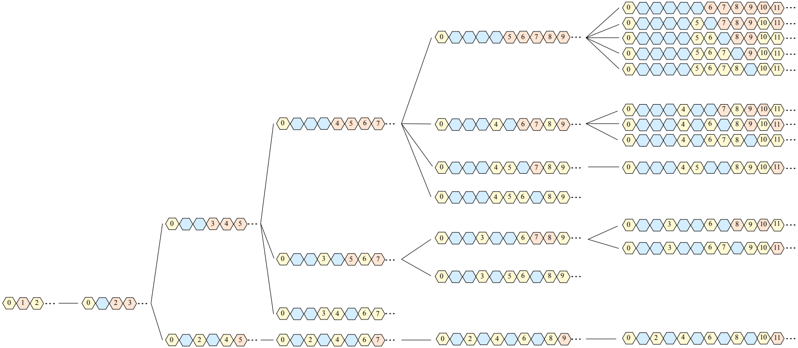
- the gap set $Gap(H) = \mathbb{N} \setminus H$
- genus $g(H) = |\mathbb{N} \setminus H|$
- Frobenius number $F(H) = \max \mathbb{Z} \setminus H$. (Note that $F(\mathbb{N}) = -1$).

Oultine of the talk:

1. The semigroup tree
2. The type
3. Big data...and questions
4. t close to g -a theorem
5. More questions

The tree of numerical semigroups

- Rosales (1991): places the numerical semigroup H in a tree with $\mathbb{N}$ as the root and the distance from H to the root is $g(H)$
- If H is a numerical semigroup with $g(H) > 0$, its *parent* is $H' = H \cup \{F(H)\}$ and $g(H') = g(H) - 1$.
- The *descendents* of H are $H \setminus \{a\}$, where a is a minimal generator for H and $a > F(H)$.
- For $H = \langle 3, 7, 8 \rangle = \{0, -, -3, -, -, 6, 7, 8, \ldots\}$ with $F(H) = 5$
  the *descendents* are $H_1 = H \setminus \{7\} = \langle 3, 8, 10 \rangle$ and $H_2 = H \setminus \{8\} = \langle 3, 7, 11 \rangle$, with $F(H_1) = 7, F(H_2) = 8$.
  The *parent* of H is $H' = H \cup \{5\} = \langle 3, 5, 7, 8 \rangle$.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The tree of numerical semigroups

Question

How fast does the semigroup tree grow?

- Set $n(g)$=the number of numerical semigroups of genus g.

Conjecture (Bras Amoros, 2008)

1. $n(g) \geq n(g-1) + n(g-2)$ *for all* $g \geq 2$
2. *(weaker)* $n(g) \geq n(g-1)$ *for* $g \geq 1$.

Theorem (Zhai, 2012)

$$\lim_{g \to \infty} \frac{n(g)}{\varphi^g} = S,$$

with $S \geq 3.78$*, and* φ *the golden ratio.*

The type of a numerical semigroup

- The *pseudo-Frobenius numbers* of H
  $PF(H) = \{x \in \mathbb{Z} \setminus H : x + h \in H, \text{ for all } 0 \neq h \in H\}$.
- the *type* of H is $t(H) = |PF(H)|$.
- Facts:
  1. $PF(H) \subseteq Gaps(H)$
  2. $PF(H) = Maximals_{\leq_H}(\mathbb{Z} \setminus H)$, where $x \leq y$ iff $y - x \in H$
  3. $1 \leq t(H) \leq g(H)$
  4. $t(H) = 1$ when H is symmetric
  5. $t(H) = g(H)$ when $H = \langle c, c+1, \ldots, 2c-1 \rangle$. Then
     $t(H) = g(H) = F(H) = c - 1$.
  6. If $H' = H \cup \{F(H)\}$ then $t(H) \geq t(H') - 1$, with equality iff
     $H' = \langle c, c+1, \ldots, 2c-1 \rangle$.

Genus vs. type

type / genus	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
1	1																																
2	1	1																															
3	2	1	1																														
4	3	2	1	1																													
5	3	4	3	1	1																												
6	6	6	6	3	1	1																											
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9	15	21	35	21	14	7	3	1	1																								
10	20	32	56	40	32	12	7	3	1	1																							
11	18	47	94	69	60	28	15	7	3	1	1																						
12	36	63	136	132	108	57	33	15	7	3	1	1																					
13	44	88	217	210	202	111	70	32	15	7	3	1	1																				
14	45	136	322	343	361	219	137	68	35	15	7	3	1	1																			
15	83	159	470	565	619	405	289	130	75	35	15	7	3	1	1																		
16	109	227	672	897	1039	754	533	289	152	72	35	15	7	3	1	1																	
17	101	329	1025	1309	1730	1364	1026	546	328	147	78	35	15	7	3	1	1																
18	174	413	1376	2090	2778	2392	1879	1103	651	312	159	78	35	15	7	3	1	1															
19	246	562	1938	3067	4441	4087	3426	2122	1298	626	358	153	78	35	15	7	3	1	1														
20	227	812	2810	4422	6967	6909	6024	4043	2523	1307	709	342	161	78	35	15	7	3	1	1													
21	420	948	3798	6706	10613	11318	10684	7399	4871	2617	1474	684	361	161	78	35	15	7	3	1	1												
22	546	1331	5179	9644	16265	18321	18134	13642	9226	5168	2956	1440	733	360	161	78	35	15	7	3	1	1											
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26	1121	4207	18579	37735	77889	109057	132168	125194	102947	69067	43209	23899	12970	6457	3323	1585	757	367	161	78	35	15	7	3	1	1							
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33	8925	23282	133674	355362	882069	1625034	2645417	3435192	3894557	3668138	3016976	2127885	1367186	794338	443863	233104	121816	60317	30733	14631	7214	3436	1632	757	367	161	78	35	15	7	3	1	1

Genus vs. type

- Denote $n(g,t) =$ the number of numerical semigroups of genus g and type t.

Playing with GAP we noticed:

- $n(g,g) = 1$ for all $g \geq 1$
- $n(g,g-1) = 1$ for all $g \geq 2$
- $n(g,g-2) = 3$ for all $g \geq 5$
- $n(g,g-3) = 7$ for all $g \geq 8$
- $n(g,g-4) = 15$ for all $g \geq 11$...
- It seems that for fixed ℓ:
  $n(g,g-\ell) =$ constant for $g \geq 3\ell - 1$.

Stable values $n(g, g-\ell)$

Theorem (CRS, 2026)

Let $g \geq t > 0$ integers with $t \geq (2g-1)/3$ (i.e. $g \leq (3t+1)/2$).

1. *If H has genus g and type t, then*
$$S = \{0\} \cup \{x+1 : 0 \neq x \in H\}$$
is a numerical semigroup of genus $g+1$ and type $t+1$.
2. *If S has genus $g+1$ and type $t+1$, then*
$$H = \{0\} \cup \{x-1 : 0 \neq x \in S\}$$
is a numerical semigroup of genus g and type t.

Corollary

- *$n(g, g-\ell)$ is constant for $g \geq 3\ell - 1$.*
- *$n(g, g-\ell) \geq \ell^2 - 3\ell + 10$ for $\ell \geq 6$.*

Part 1 of the Corollary was proved also by D. Singhal (IJAC, 2021).

Gap vectors

Proposition (Nari)

For any numerical semigroup H

$$t(H) + F(H) \leq 2g(H).$$

So, if we denote $\ell = g(H) - t(H)$ one has

$$\{1, 2, \dots, g - \ell\} \subseteq Gaps(H) \subseteq \{1, 2, \dots, g + \ell\}.$$

We defined the *gap vector* of H as $v(H) = (v_1, \dots, v_{2\ell})$

$$v_i = \begin{cases} 1 & \text{if } g - \ell + i \in Gap(H) \\ 0 & \text{else,} \end{cases}$$

where ℓ entries are equal to 1 and ℓ are equal to 0.

Cotype

For any $g \geq \ell > 0$ and $v = (v_1, \ldots, v_{2\ell}) \in \{0,1\}^{2\ell}$ set

$$\begin{aligned} \mathcal{G}_{g,v} &= \{1, \ldots, g-\ell\} \cup \{g - \ell + i : 1 \leq i \leq 2\ell, v_i = 1\} \\ H_{g,v} &= \mathbb{N} \setminus \mathcal{G}_{g,v}. \end{aligned}$$

Proposition

If $g \geq 3\ell - 1$ and $v \in \{0,1\}^{2\ell}$ has ℓ entries equal to 1, then

1. *$H_{g,v}$ is a numerical semigroup of genus g.*
2. *$t(H_{g,v}) = g - \mathit{cotype}(v)$, where we define*

$$\mathit{cotype}(v) = |\{j - i : 1 \leq i < j \leq 2\ell, v_i = 0, v_j = 1\}|$$

Example

1. $\mathit{cotype}(1, \ldots, 1, 0, \ldots, 0) = |\emptyset| = 0$, hence $H_{g,v} = \langle g+1, \ldots, 2g+1 \rangle$ has type $= g$
2. $\mathit{cotype}(0, \ldots, 0, 1, \ldots, 1) = |\{1, 2, \ldots, \ell, 2\ell - 1\}| = 2\ell - 1$, hence $H_{g,v}$ has type $= g - 2\ell + 1$.

Stable gap vectors

According to our Main Theorem, numerical semigroups H with type g and type $g-\ell$ have the same gap vector once $g \geq 3\ell - 1$.
So, if one is to find these semigroups, one needs to describe these *stable gap vectors* that may occur.

• For $\ell = 1$ $\quad v = (1, 0)$, and $n(g, g-1) = 1$ for $g \geq 2$.
• For $\ell = 2$ $\quad v \in \{(1, 0, 0, 1), (0, 1, 1, 0), (0, 1, 0, 1)\}$, and $n(g, g-2) = 3$ for $g \geq 5$.
• For $\ell = 3$ $\quad v \in \{(1, 1, 0, 0, 0, 1), (1, 0, 1, 0, 0, 1), (1, 0, 0, 1, 1, 0)$ $(0, 1, 1, 1, 0, 0), (0, 1, 1, 0, 1, 0), (0, 1, 1, 0, 0, 1), (0, 1, 0, 1, 0, 1)\}$, and $n(g, g-3) = 7$ for $g \geq 8$.

Questions/Unimodality

Question

Is it true that the sequence

$$n(g,1), n(g,2), \ldots, n(g,g)$$

is unimodal for all g?

Question

Is it true that for any $t > 1$ and all $g > 0$ one has

$$n(g,t) < n(g+1,t)?$$

- We verified these inequalities with GAP for $g \leq 33$.
- Note that $n(25,1) = 1182 < 1121 = n(26,1)$, so the second question needs $t > 1$.

Thank you!